

ECHS FRAUD FORENSICS

BEHAVIORAL DETECTION PATTERNS & REAL DATA EVIDENCE

This document bridges theoretical audit patterns with explicit statistical findings derived from the Module 20 forensic database extraction (19.8 Million cases / ₹3,612 Cr exposure).

PATTERN 1Corporate Hospital Overbilling

CORE CONCEPT

Instead of relying on percentage-based rejection rates (which artificially flag small clinics), this pattern targets massive corporate chains by **absolute leakage volume** to identify the largest sinks of ECHS funds.

BYPASS MECHANISM

Auditors rank facilities by the total Rupee value deducted across all claims, isolating corporate entities driving systemic leakage.

REAL DATA EVIDENCE

The **Vijay Hospital (ID 3149)** exhibited an anomalous 34.00% deduction rate. Concurrently, the **Park Hospital Chain** (Gurgaon, Chowkhandi, Kailash) emerged as the largest corporate target, requiring an immediate chain-level empanelment review.

PATTERN 2Macro Analytics: Annual, Regional & Demographics

CORE CONCEPT

This mega-pattern evaluates systemic structural leakage. It identifies year-over-year inflation, geographic hotspots, and targeted patient demographics.

BYPASS MECHANISM

By aggregating millions of records along these three dimensions, the system flags entire segments of the ECHS program operating outside normal statistical bounds.

REAL DATA EVIDENCE

Chennai Region 6 was flagged with a severe geographic anomaly (**19.13%** deduction rate vs 10% average). Demographically, the **'Wife'** category exhibited an elevated rejection rate of **15.59%** , exposing targeted upcoding in female-specific IPD procedures.

PATTERN 3 Itemized Procedure Deviations

CORE CONCEPT

Detects the unbundling of surgical packages and unjustified pharmacy inflation.

BYPASS MECHANISM

The system runs text-analytics on medical auditor remarks. It flags when a hospital attempts to bill separately for oxygen, ICU, or equipment already covered in the fixed package.

REAL DATA EVIDENCE

Auditors heavily cited **'Package Double-Billing'** and **'High-End Antibiotic Abuse'**. E.g., multiple high-end antibiotics were administered in massive, clinically unjustified doses solely to inflate the pharmacy bill.

PATTERN 4 Length of Stay (LoS) Bed-Blocking

CORE CONCEPT

Detects hospitals deliberately keeping patients admitted for unnecessarily long durations (>10 days) without clinical justification.

BYPASS MECHANISM

The script calculates the gap between Admission and Discharge dates for routine ailments, identifying pure room-rent and nursing charge inflation.

REAL DATA EVIDENCE

The system isolated multiple cases where routine ailments were stretched to **>15 days**, converting standard observation cases into highly lucrative, extended IPD stays.

PATTERN 5 Ping-Pong Admissions (Split-Package Fraud)

CORE CONCEPT

Identifies hospitals circumventing fixed package duration limits by discharging and immediately readmitting patients within 48 hours.

BYPASS MECHANISM

The patient is never physically discharged. The hospital merely closes their file and opens a new one (Ping-Ponging) to trigger a fresh package rate.

REAL DATA EVIDENCE

The extraction detected numerous readmissions with a literal **0 Days Gap**. This is definitive proof of paperwork manipulation to split a single continuous stay into two separate claims.

PATTERN 6

Weekend Admission Surge (The Friday Hustle)

CORE CONCEPT

Exploiting the physical absence of ECHS verifiers during the weekends.

BYPASS MECHANISM

Analyzes the day-of-the-week distribution for admissions. A massive spike strictly on Fridays, Saturdays, and Sundays indicates the hospital is pushing through unnecessary or fake admissions while authorities are off-duty.

REAL DATA EVIDENCE

A high concentration of 'Suspicious Weekend Admission Spikes' was flagged, pinpointing facilities orchestrating their highest IPD intake exactly when oversight is minimal.

PATTERN 7

Doctor Cloning (The Superman Surgeon)

CORE CONCEPT

Detects physical impossibility by flagging a single 'Treating Doctor' attached to an absurd number of surgeries in a single day.

BYPASS MECHANISM

If a surgeon bills for >15 complex procedures in 24 hours, the hospital is bulk-billing under one ID or hiding the true physicians.

REAL DATA EVIDENCE

Hospitals are bypassing the system using generic garbage text: 'CMO' (79 surgeries in one day at Sarvodaya Hospital), 'ECHS' (80 at Max Super Speciality). Even dentists like Dr. Naveen Garg logged an impossible 80 procedures in a single day.

PATTERN 8

Threshold Avoiding (The ₹99k Trick)

CORE CONCEPT

Hospitals intentionally manipulating the final bill amount to avoid senior scrutiny.

BYPASS MECHANISM

If an automatic CFA approval threshold is set at ₹1,00,000, hospitals will intentionally bill exactly ₹99,000 or ₹99,999 so the claim slips through the automated system unnoticed.

REAL DATA EVIDENCE

The algorithm successfully flagged a high volume of 'Trick Bills' sitting uniformly tight against the automated approval ceiling, demonstrating a calculated evasion of senior officer review.